Test sheet: the axis of a mirror-symmetric mark

The mark-axis rule (stage 4) puts a symmetric mark's anchor X on its axis, and takes the axis to be the centre of the mark's ink bounds, with the slant removed. The open decision in the report ("Mark anchor model", "Open decisions") suggested that a more precise axis would be the line about which the outline best matches its mirror image, or, for the tilde, zigzag and vertical tilde, the centre about which it best matches its rotation by 180°, since their terminals could push the bounds off centre.

This sheet measures that axis for every mark in `SYMMETRIC_ABOVE` (except the combining letters, which are not mirror-symmetric and have their own measures in `AXIS_MEASURE`) and `SYMMETRIC_BELOW`, and the symmetric `.cap` forms, in the four patched fonts. Method: the outline, de-slanted at the style's declared slant, is scanned into ink intervals every half unit of height; the mirror axis is the x that maximizes the overlap of the outline with its mirror image (searched to ± 30 units of the bbox centre, refined to 0.025); the rotation centre is the point that maximizes the overlap with the outline rotated by 180°. For each mark the symmetry with the smaller mismatch is used. Δ is the symmetry axis minus the bbox centre: how far the anchor would move.

Result

Across all 204 glyphs, the largest $|\Delta|$ is 2.2 units (semibold italic `breve.cap`); 188 of 204 are under 1 unit. The tilde-like marks (tilde, tilde below, perispomeni, vertical tilde, zigzag, almost equal to, not tilde, homothetic) match their 180° rotation about a centre within 1.3 units of the bbox centre; their mirror axis is meaningless (the search hits its ± 30 limit, mismatch about 50%). Not tilde (U+034A) and homothetic (U+034B) match neither symmetry exactly (mismatch about 10%), since they combine a tilde with straight strokes, but the rotation centre is still within 1 unit. The italic and semibold italic dots (U+0307, U+0308) are skewed ovals: their mirror mismatch is 13%, their rotation mismatch small, and both axes are on the bbox centre.

Decision to make: none of the differences reaches 5 units; the biggest is 2.2, and the anchors are integers. Recommendation: keep the bbox centre, and move the item from "Open decisions" to "Decided: left as they are" with these figures.

Δ in units (m: mirror; r: 180° rotation; mismatch in %, the share of ink not covered by its image)

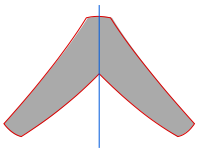
	mark	Regular	Italic	Semibold	Semibold italic
above	U+0302 circumflex	+0.2 m (1)	-0.6 m (1)	-0.2 m (1)	-0.3 m (1)
above	U+0303 tilde	-0.5 r (2)	-0.5 r (2)	-1.2 r (2)	-0.5 r (3)
above	U+0304 macron	-0.1 r (0)	-0.4 r (0)	+0.0 r (0)	+0.1 r (0)
above	U+0306 breve	+0.1 m (2)	+0.5 m (1)	-0.2 m (1)	-0.1 m (1)
above	U+0307 dot above	+0.0 r (1)	-0.1 r (1)	+0.0 m (0)	-0.1 r (0)
above	U+0308 dieresis	+0.0 m (0)	-0.1 r (1)	+0.0 m (-0)	-0.1 r (1)
above	U+030A ring above	+0.0 m (0)	+0.0 r (1)	-0.4 m (1)	+0.4 m (1)
above	U+030C caron	+0.0 m (1)	+0.3 m (0)	-0.1 m (1)	+0.1 m (1)
above	U+030D vertical line above	+0.0 r (0)	-0.1 r (1)	+0.0 m (0)	+0.0 r (0)
above	U+030E double vertical line above	+0.5 m (1)	+0.4 r (1)	+0.0 m (0)	+0.0 m (1)
above	U+0310 candrabindu	+0.0 m (1)	+0.2 m (2)	-0.3 m (3)	-0.2 m (3)
above	U+0311 inverted breve	-0.1 m (2)	-0.5 m (1)	+0.2 m (1)	-0.1 m (1)
above	U+033D x above	+0.2 m (1)	+0.1 r (1)	+0.2 m (1)	+0.1 r (1)
above	U+033E vertical tilde	+1.3 r (3)	+0.8 r (3)	+1.3 r (3)	+0.8 r (3)
above	U+033F double overline	-0.1 r (1)	-0.2 r (1)	-0.1 r (1)	-0.2 r (1)
above	U+0342 perispomeni	-0.5 r (2)	-0.6 r (3)	-1.2 r (2)	-1.1 r (3)
above	U+0346 bridge above	+0.0 m (1)	+0.2 m (1)	+0.0 m (1)	+0.1 m (1)
above	U+034A not tilde above	+0.0 r (10)	+0.1 r (11)	+0.0 r (9)	+0.0 r (10)
above	U+034B homothetic above	-0.5 r (11)	-0.5 r (12)	-0.7 r (10)	-0.3 r (14)
above	U+034C almost equal to above	+1.1 r (2)	+0.8 r (3)	+1.1 r (2)	+1.1 r (2)
above	U+0352 fermata	+0.0 m (1)	+0.1 m (2)	+0.0 m (1)	-0.1 m (1)
above	U+035B zigzag above	+0.0 r (1)	+0.3 r (1)	+0.5 r (1)	+0.3 r (1)
above	`circumflex.cap`	-0.7 m (1)	-0.4 m (1)	-0.8 m (1)	-1.4 m (4)
above	`tildecomb.cap`	-0.5 r (2)	-0.5 r (2)	-1.2 r (2)	-0.7 r (2)
above	`breve.cap`	-0.5 m (1)	+1.6 m (2)	-0.2 m (1)	+2.2 m (2)
above	`dotaccent.cap`	+0.0 m (0)	+0.2 m (0)	+0.0 m (0)	-0.1 r (0)
above	`dieresis.cap`	+0.0 m (0)	+0.4 r (1)	+0.1 m (0)	-0.1 r (1)
above	`caron.cap`	+0.5 m (1)	+0.1 m (1)	+0.5 m (1)	+1.1 m (4)
above	`breveinvertedcmb.cap`	+0.5 m (1)	-0.1 m (2)	+0.2 m (1)	-2.0 m (2)
below	U+031F plus sign below	+0.5 r (1)	+0.1 r (1)	+0.1 r (0)	-0.1 r (0)
below	U+0320 minus sign below	-0.1 r (0)	-0.5 r (0)	-0.2 r (0)	-0.1 r (0)
below	U+0323 dot below	+0.0 m (0)	+0.0 r (1)	+0.1 m (0)	-0.1 m (0)
below	U+0324 diaeresis below	+0.0 m (-0)	+0.0 r (1)	+0.0 r (0)	-0.3 m (0)
below	U+0325 ring below	+0.0 m (0)	+0.2 r (1)	-0.3 m (1)	+0.1 m (0)
below	U+0329 vertical line below	+0.0 r (0)	+0.1 r (0)	+0.0 m (0)	+0.1 r (1)
below	U+032A bridge below	+0.0 m (1)	+0.1 m (1)	+0.0 m (1)	+0.1 m (1)
below	U+032B inverted double arch below	-0.0 m (0)	+0.2 m (1)	+0.0 m (1)	-0.1 m (1)
below	U+032C caron below	-0.2 m (1)	+0.2 m (0)	-0.2 m (1)	+0.0 m (1)
below	U+032D circumflex below	+0.1 m (1)	-0.5 m (1)	+0.1 m (1)	+0.4 m (1)

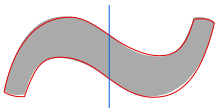
	mark	Regular	Italic	Semibold	Semibold italic
below	U+032E breve below	+0.1 m (2)	+0.0 m (2)	-0.2 m (1)	+0.1 m (1)
below	U+032F inverted breve below	-0.1 m (2)	-0.2 m (2)	+0.1 m (1)	-0.1 m (1)
below	U+0330 tilde below	-0.5 r (2)	-0.7 r (3)	-1.2 r (2)	-1.0 r (3)
below	U+0331 macron below	-0.1 r (0)	-0.2 r (0)	-0.1 r (0)	+0.0 r (0)
below	U+0333 double low line	-0.1 r (1)	+0.2 r (1)	-0.1 r (1)	-0.2 r (1)
below	U+033A inverted bridge below	+0.0 m (1)	-0.2 m (1)	+0.0 m (1)	+0.0 m (1)
below	U+033B square below	+0.0 m (0)	+0.0 r (1)	+0.0 m (0)	+0.0 m (0)
below	U+033C seagull below	+0.0 m (0)	-0.9 m (1)	+0.0 m (0)	-0.4 m (1)
below	U+0347 equals sign below	-0.1 r (1)	+0.2 r (1)	-0.1 r (1)	-0.2 r (1)
below	U+0348 double vertical line below	+0.0 r (0)	-0.2 r (1)	+0.0 r (0)	+0.1 m (0)
below	U+034D left right arrow below	+0.1 m (2)	+0.1 m (3)	+0.1 m (2)	+0.3 m (2)
below	U+034E upwards arrow below	+0.5 m (2)	+0.4 m (2)	+0.5 m (2)	+0.5 m (2)

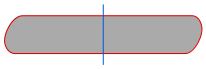
The marks with their images

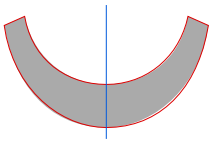
Grey: the mark's outline, de-slanted. Red: its mirror image about the bbox axis (blue), or its rotation by 180° about the point on the bbox axis at the best height, for the marks marked r. Where red follows grey, the bbox axis is the symmetry axis. Scale 0.1 mm per unit, so 1 unit of Δ would show as a red line 0.2 mm off the grey edge on each side.

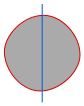
Regular

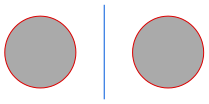

U+0302 uni0302: Δ +0.2 m

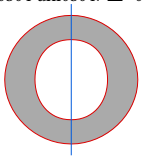

U+0303 tildecomb: Δ -0.5 r

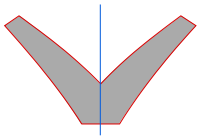

U+0304 uni0304: Δ -0.1 r

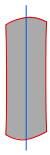

U+0306 uni0306: Δ +0.1 m

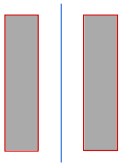

U+0307 uni0307: Δ +0.0 r

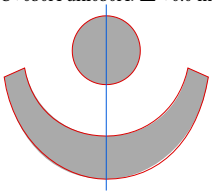

U+0308 uni0308: Δ +0.0 m

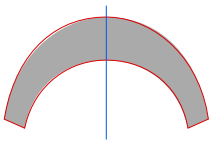

U+030A uni030A: Δ +0.0 m

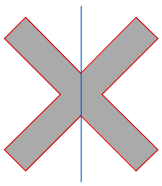

U+030C uni030C: Δ +0.0 m


U+030D uni030D: Δ +0.0 r

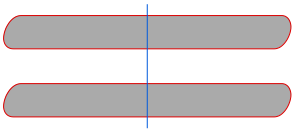

U+030E uni030E: Δ +0.5 m

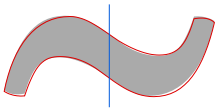

U+0310 uni0310: Δ +0.0 m

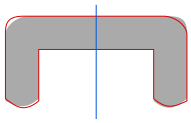

U+0311 uni0311: Δ -0.1 m

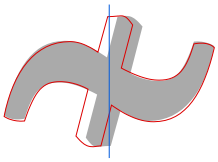

U+033D uni033D: Δ +0.2 m

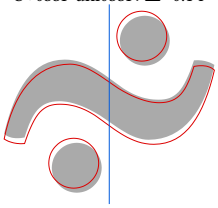

U+033E uni033E: Δ +1.3 r

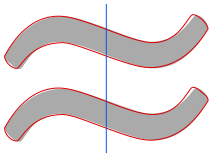

U+033F uni033F: Δ -0.1 r

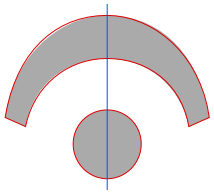

U+0342 uni0342: Δ -0.5 r

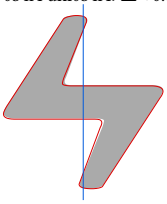

U+0346 uni0346: Δ +0.0 m

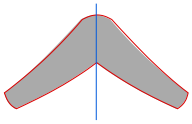

U+034A uni034A: Δ +0.0 r

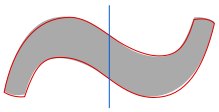

U+034B uni034B: Δ -0.5 r

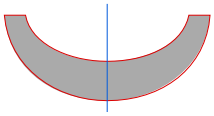

U+034C uni034C: Δ +1.1 r

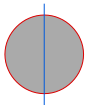

U+0352 uni0352: Δ +0.0 m

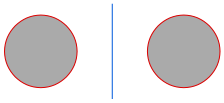

U+035B uni035B: Δ +0.0 r

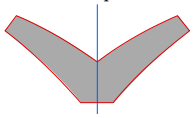

circumflex.cap: Δ -0.7 m

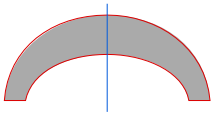

tildecomb.cap: Δ -0.5 r

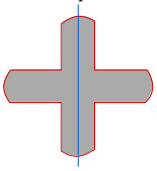

breve.cap: Δ -0.5 m


dotaccent.cap: Δ +0.0 m

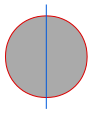

dieresis.cap: Δ +0.0 m

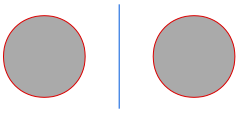

caron.cap: Δ +0.5 m

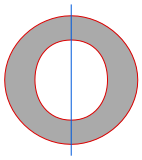

breveinvertedcmb.cap: Δ +0.5 m

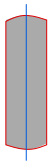

U+031F uni031F: Δ +0.5 r

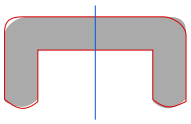

U+0320 uni0320: Δ -0.1 r

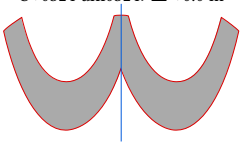

U+0323 dotbelowcomb: Δ +0.0 m

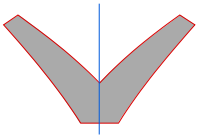

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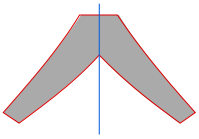

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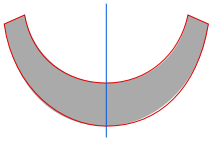

U+0329 uni0329: Δ +0.0 r

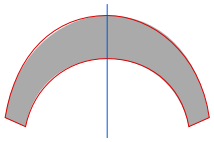

U+032A uni032A: Δ +0.0 m


U+032B uni032B: Δ -0.0 m

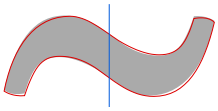

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U+032D uni032D: Δ +0.1 m

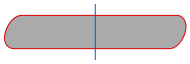

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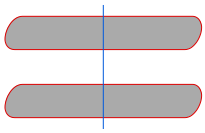
U+032F uni032F: Δ -0.1 m



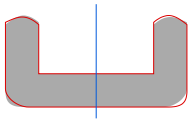
U+0330 uni0330: Δ -0.5 r



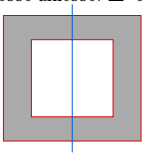
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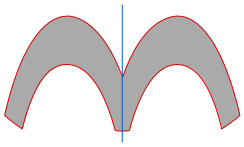
U+0333 uni0333: Δ -0.1 r



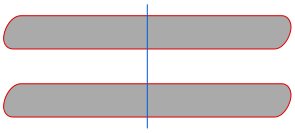
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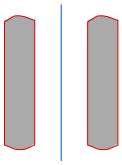
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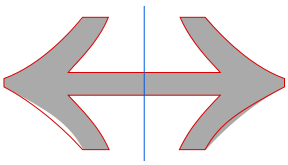
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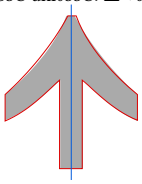
U+0347 uni0347: Δ -0.1 r



U+0348 uni0348: Δ +0.0 r

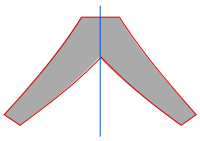


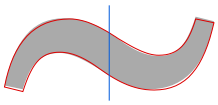
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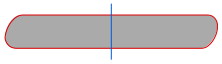


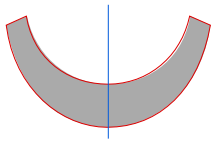
U+034E uni034E: Δ +0.5 m

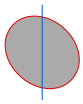
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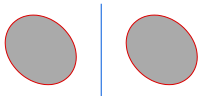

U+0302 uni0302: Δ -0.6 m

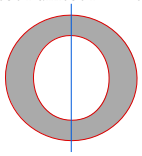

U+0303 tildecomb: Δ -0.5 r

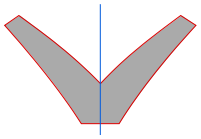

U+0304 uni0304: Δ -0.4 r

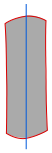

U+0306 uni0306: Δ +0.5 m

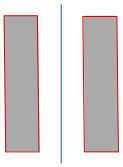

U+0307 uni0307: Δ -0.1 r

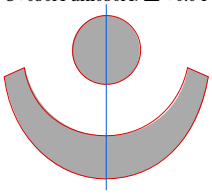

U+0308 uni0308: Δ -0.1 r

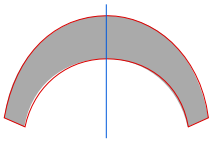

U+030A uni030A: Δ +0.0 r

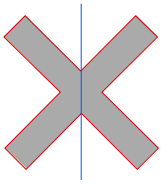

U+030C uni030C: Δ +0.3 m


U+030D uni030D: Δ -0.1 r

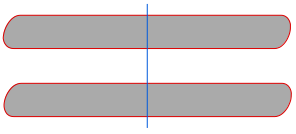

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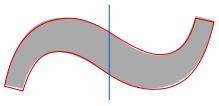

U+0310 uni0310: Δ +0.2 m

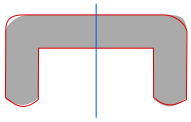

U+0311 uni0311: Δ -0.5 m

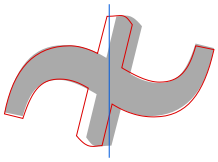

U+033D uni033D: Δ +0.1 r

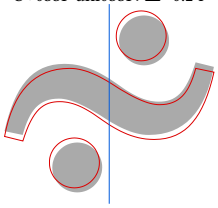

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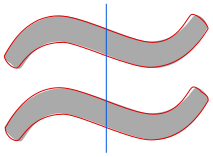

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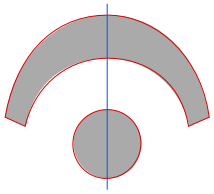

U+0342 uni0342: Δ -0.6 r

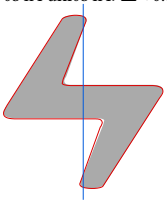

U+0346 uni0346: Δ +0.2 m

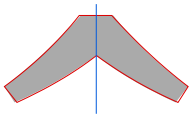

U+034A uni034A: Δ +0.1 r

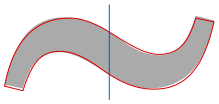

U+034B uni034B: Δ -0.5 r

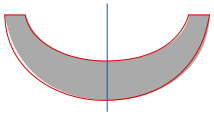

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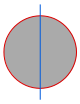

U+0352 uni0352: Δ +0.1 m

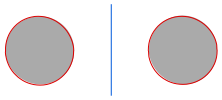

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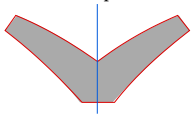

circumflex.cap: Δ -0.4 m

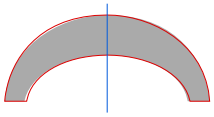

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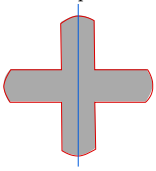

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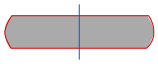

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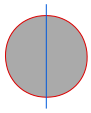

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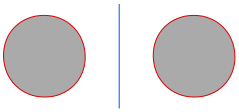

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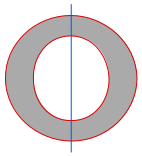

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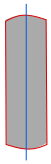

U+031F uni031F: Δ +0.1 r

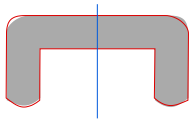

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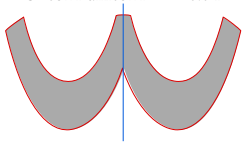

U+0323 dotbelowcomb: Δ +0.0 r

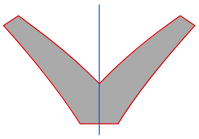

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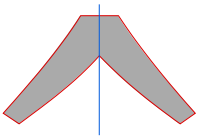

U+0325 uni0325: Δ +0.2 r

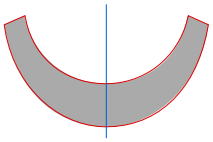

U+0329 uni0329: Δ +0.1 r

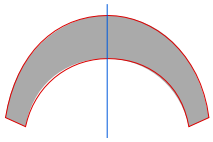

U+032A uni032A: Δ +0.1 m


U+032B uni032B: Δ +0.2 m

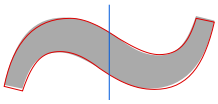

U+032C uni032C: Δ +0.2 m


U+032D uni032D: Δ -0.5 m

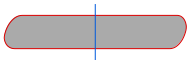

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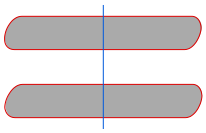
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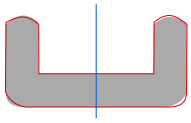
U+0330 uni0330: Δ -0.7 r



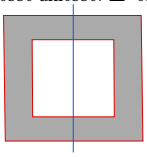
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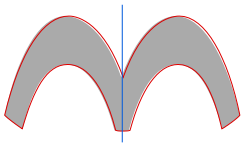
U+0333 uni0333: Δ +0.2 r



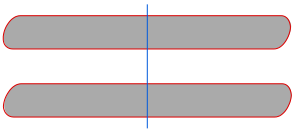
U+033A uni033A: Δ -0.2 m



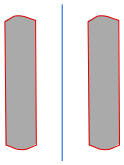
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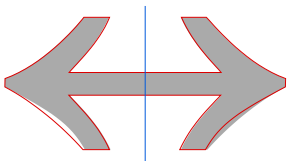
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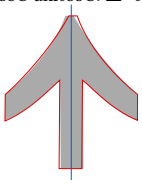
U+0347 uni0347: Δ +0.2 r



U+0348 uni0348: Δ -0.2 r

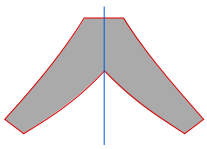


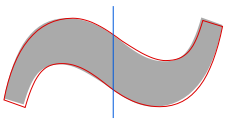
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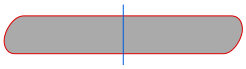


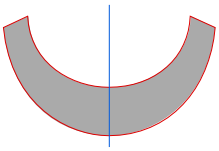
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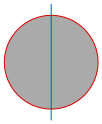
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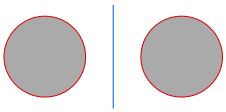

U+0302 uni0302: Δ -0.2 m

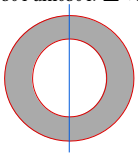

U+0303 tildecomb: Δ -1.2 r

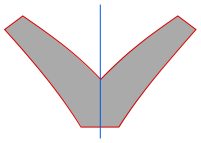

U+0304 uni0304: Δ +0.0 r

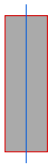

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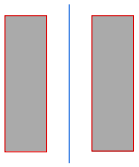

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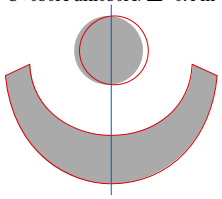

U+0308 uni0308: Δ +0.0 m

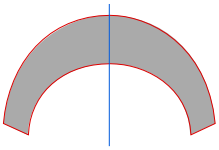

U+030A uni030A: Δ -0.4 m

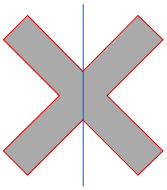

U+030C uni030C: Δ -0.1 m

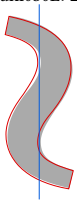

U+030D uni030D: Δ +0.0 m

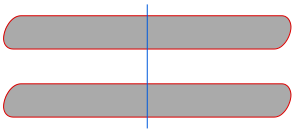

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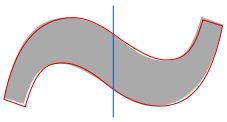

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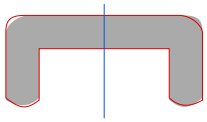

U+0311 uni0311: Δ +0.2 m

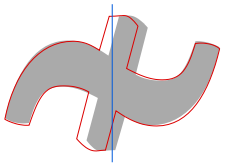

U+033D uni033D: Δ +0.2 m

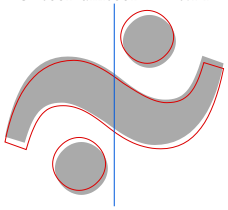

U+033E uni033E: Δ +1.3 r

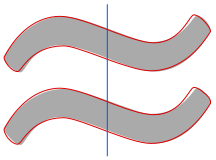

U+033F uni033F: Δ -0.1 r

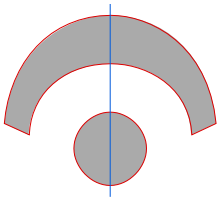

U+0342 uni0342: Δ -1.2 r

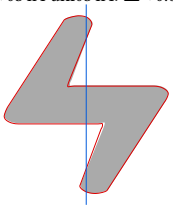

U+0346 uni0346: Δ +0.0 m

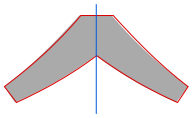

U+034A uni034A: Δ +0.0 r

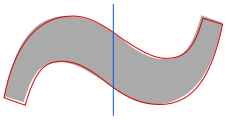

U+034B uni034B: Δ -0.7 r

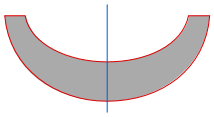

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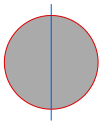

U+0352 uni0352: Δ +0.0 m

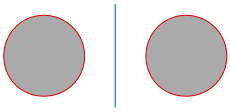

U+035B uni035B: Δ +0.5 r

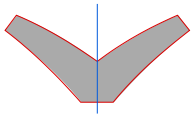

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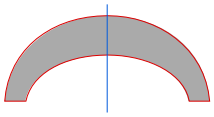

tildecomb.cap: Δ -1.2 r

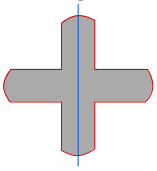

breve.cap: Δ -0.2 m

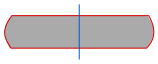

dotaccent.cap: Δ +0.0 m

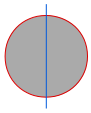

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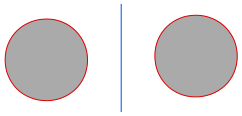

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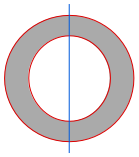

breveinvertedcmb.cap: Δ +0.2 m

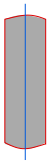

U+031F uni031F: Δ +0.1 r

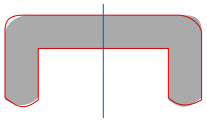

U+0320 uni0320: Δ -0.2 r

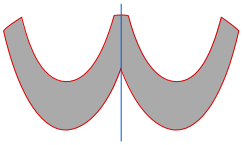

U+0323 dotbelowcomb: Δ +0.1 m

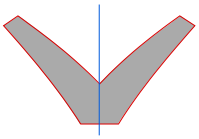

U+0324 uni0324: Δ +0.0 r

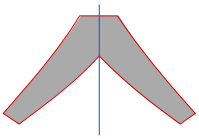

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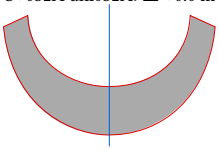

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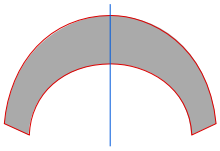

U+032A uni032A: Δ +0.0 m


U+032B uni032B: Δ +0.0 m

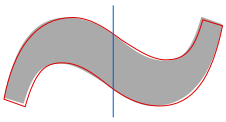

U+032C uni032C: Δ -0.2 m


U+032D uni032D: Δ +0.1 m


U+032E uni032E: Δ -0.2 m



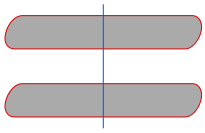
U+032F uni032F: Δ +0.1 m



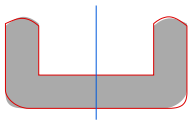
U+0330 uni0330: Δ -1.2 r



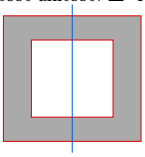
U+0331 uni0331: Δ -0.1 r



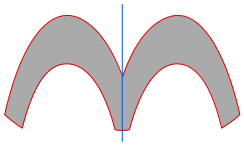
U+0333 uni0333: Δ -0.1 r



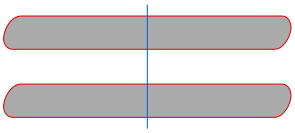
U+033A uni033A: Δ +0.0 m



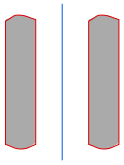
U+033B uni033B: Δ +0.0 m



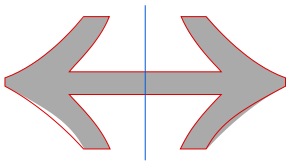
U+033C uni033C: Δ +0.0 m



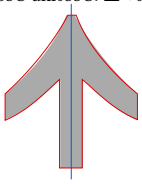
U+0347 uni0347: Δ -0.1 r



U+0348 uni0348: Δ +0.0 r

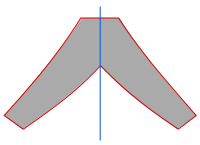


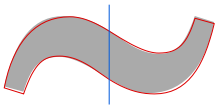
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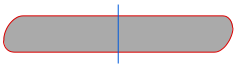


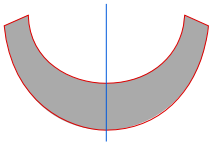
U+034E uni034E: Δ +0.5 m

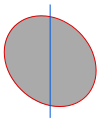
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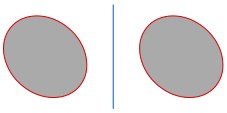

U+0302 uni0302: Δ -0.3 m

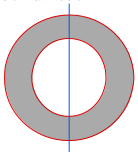

U+0303 tildecomb: Δ -0.5 r

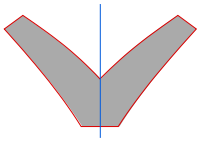

U+0304 uni0304: Δ +0.1 r

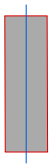

U+0306 uni0306: Δ -0.1 m

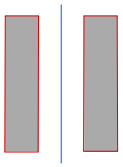

U+0307 uni0307: Δ -0.1 r

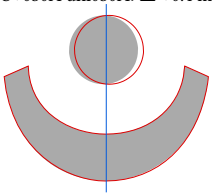

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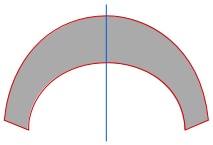

U+030A uni030A: Δ +0.4 m

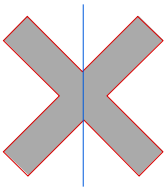

U+030C uni030C: Δ +0.1 m

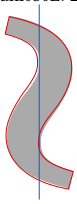

U+030D uni030D: Δ +0.0 r

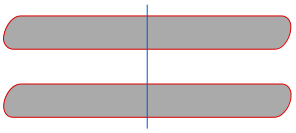

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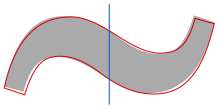

U+0310 uni0310: Δ -0.2 m

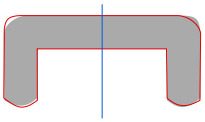

U+0311 uni0311: Δ -0.1 m

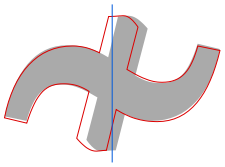

U+033D uni033D: Δ +0.1 r

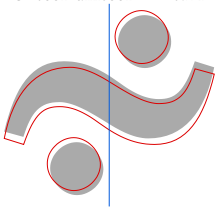

U+033E uni033E: Δ +0.8 r

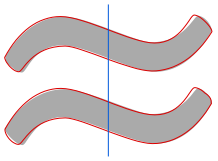

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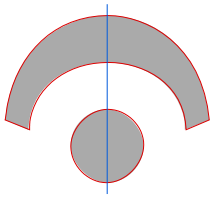

U+0342 uni0342: Δ -1.1 r

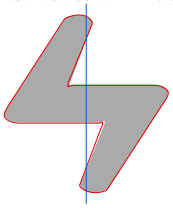

U+0346 uni0346: Δ +0.1 m

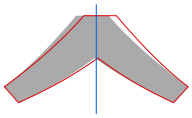

U+034A uni034A: Δ +0.0 r

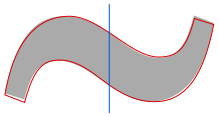

U+034B uni034B: Δ -0.3 r

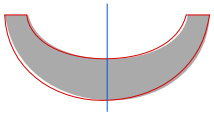

U+034C uni034C: Δ +1.1 r

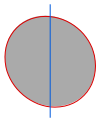

U+0352 uni0352: Δ -0.1 m

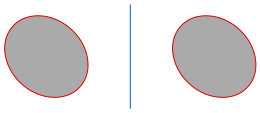

U+035B uni035B: Δ +0.3 r

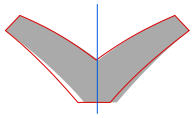

circumflex.cap: Δ -1.4 m

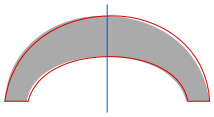

tildecomb.cap: Δ -0.7 r

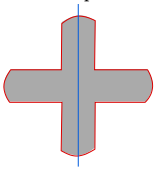

breve.cap: Δ +2.2 m

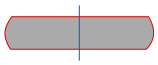

dotaccent.cap: Δ -0.1 r

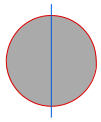

dieresis.cap: Δ -0.1 r

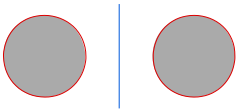

caron.cap: Δ +1.1 m

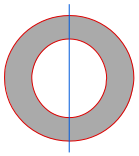

breveinvertedcmb.cap: Δ -2.0 m

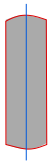

U+031F uni031F: Δ -0.1 r

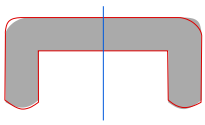

U+0320 uni0320: Δ -0.1 r

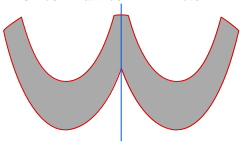

U+0323 dotbelowcomb: Δ -0.1 m

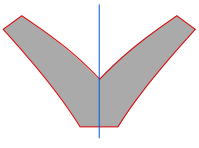

U+0324 uni0324: Δ -0.3 m

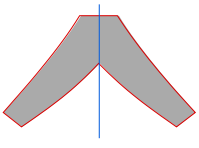

U+0325 uni0325: Δ +0.1 m

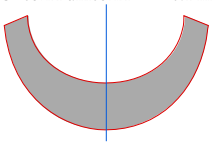

U+0329 uni0329: Δ +0.1 r

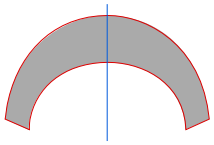

U+032A uni032A: Δ +0.1 m


U+032B uni032B: Δ -0.1 m

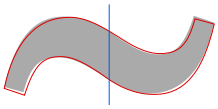

U+032C uni032C: Δ +0.0 m


U+032D uni032D: Δ +0.4 m

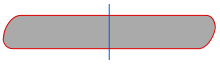

U+032E uni032E: Δ +0.1 m



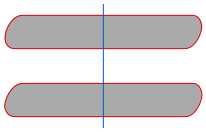
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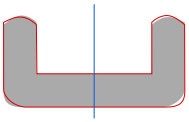
U+0330 uni0330: Δ -1.0 r



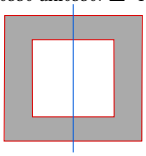
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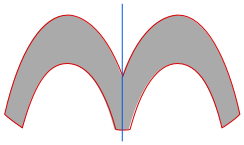
U+0333 uni0333: Δ -0.2 r



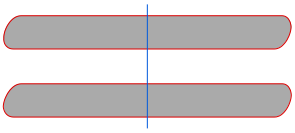
U+033A uni033A: Δ +0.0 m



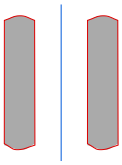
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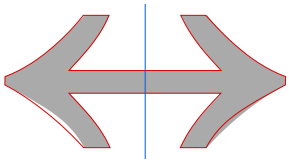
U+033C uni033C: Δ -0.4 m



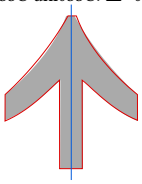
U+0347 uni0347: Δ -0.2 r



U+0348 uni0348: Δ +0.1 m



U+034D uni034D: Δ +0.3 m



U+034E uni034E: Δ +0.5 m